Hybrid comparison: identical magnitude-domain data and detector

2 paths/user | 4 paths/user | 3–7 paths/user

RSR 5 dB Uncoded BER

RSR 12 dB Uncoded BER

RSR 20 dB Uncoded BER

10^{-1} 10^{-2} 10^{-3} 10^{-4} 10^{-5} 10^{-6}

0 20 30 40 50 60

SNR (dB)

Perfect CSI
Hybrid
Same hybrid, no extra EM
Frozen latest estimator
Adaptive start + EM40
Global schedule + EM40
Classical start + EM40
Compact sparse
Zero observed errors (marker only)

Whole grid is primary; subgroups are descriptive. Zero-error markers at 10^{-6} are display markers, not estimates or bounds.